

Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Ayul, Paul
Date of birth:	11/10/1991
Subject:	030-Flight Performance&Planing
Result:	38,99 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	388	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
3	387	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	390	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
6	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
7	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	315	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
9	380	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	376	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	382	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
12	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
13	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
14	257	31.3.1	Definition of centre of gravity	0,00	1,00
15	278	31.3.3	Basic calculations of CG	0,00	2,00
16	411	32.	PERFORMANCE - AEROPLANES	1,00	1,00
17	428	32.	PERFORMANCE - AEROPLANES	1,00	1,00
18	423	32.	PERFORMANCE - AEROPLANES	0,00	1,00
19	404	32.	PERFORMANCE - AEROPLANES	0,00	1,00
20	339	32.	PERFORMANCE - AEROPLANES	0,00	1,00
21	36	32.2.5.4	Landing	1,00	1,00
22	132	32.3.3.1	Take-off	1,00	1,00
23	115	32.4.1	Take - off	0,00	1,00
24	101	32.4.1.7	Take-off climb	0,00	1,00
25	135	32.4.1.7	Take-off climb	0,00	1,00
26	15	32.4.3	Cruise	0,00	1,00
27	202	32.4.3	Cruise	0,00	1,00
28	133	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	96	32.4.3.5	Influence of variables on cruise performance	1,00	1,00
30	134	32.4.5	Descent	1,00	1,00
31	18	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
32	416	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
33	418	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
34	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
35	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
36	92	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
37	148	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	0,00	1,00
38	175	33.2.1.2	Courses and distances from en-route charts	1,00	1,00
39	30	33.5.1.1	Format of Flight Plan	0,00	1,00
40	88	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:39%				16,00	41,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Ayul, Paul

Location: Uganda Civil Aviation Authority

Your Result: 16,00 from 41,00 Points (39,02 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	4,00	16,00	25,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	2	0,00	3,00	0,00
32. PERFORMANCE - AEROPLANES	15	6,00	15,00	40,00
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	1	1,00	1,00	100,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	1	1,00	1,00	100,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	8	2,00	8,00	25,00
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	6,00	10,00	60,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	1	0,00	1,00	0,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	1,00	1,00	100,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	2	0,00	2,00	0,00
Total	40	16,00	41,00	39,02

1

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☒ **There is no relationship between CG location and indicated airspeed.**
- ☐ **Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.**
- ☐ **Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.**

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2

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☐ No; the new CG would be located at Station 118.15.

☐ Yes; the CG would remain at Station 115.8.

☒ Yes; the new CG would be located at Station 117.69.

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3

As the CG moves aft, an aircraft becomes

- ☐ less stable, yet easier to control.
- ☒ more stable and controllable as long as the aft CG is not exceeded.
- ☐ less stable and less controllable.

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4

Under which condition is a forward CG most critical?

☒ **On takeoff.**

☐ **When in an unusual attitude.**

☐ **On landing.**

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5

As the CG location is changed, recovery from a stall becomes progressively

- ☒ more difficult as the CG moves rearward.
- ☐ less difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.

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6

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 81.3.

☒ 80.6.

☐ 83.4.

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7

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☒ 968.6 pounds.

☐ 924.5 pounds.

☐ 953.6 pounds.

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8

The term 'weigh-off' as used in ballooning means to determine the

- ☒ standard weight and balance of the balloon.
- ☐ amount of fuel required for ascent to a preselected altitude.
- ☐ static equilibrium of the balloon as loaded for flight.

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9

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

☒ No; the new CG would be located at Station 116.89.

☐ Yes; the new CG would be located at Station 117.89.

☐ No; the new CG would be located at Station 116.42.

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10

What is the location of the CG if 90 pounds are removed from Station 140?

Aircraft weight 6,230 lb

CG location Station 79

☐ 78.1.

☒ 79.9.

☐ 77.9.

11

The center of gravity of an aircraft can be determined by

- ☒ dividing total moment by total weight.
- ☐ multiplying total arm by total weight.
- ☐ dividing total arm by total moments.

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What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☐ 85.1.

☒ 84.1.

☐ 84.9.

13

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☒ **Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.**
- ☐ **The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.**
- ☐ **The aft CG provides the highest stall speed and cruising speed.**

14

When an aircraft is stationary on the ground, its total weight will act vertically

- ☐ *through its centre of gravity.*
- ☒ *through the main wheels of its undercarriage assembly.*
- ☐ *through a point defined as the datum point.*
- ☐ *through its centre of pressure.*

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15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

☐ $A = B + Fb / Fa$

☒ $A = B (Fa + Fb)$

☐ $A = B \times Fa / Fb$

☐ $A = B \times Fb / Fa$

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

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16

What is the best method of speed reduction if hydroplaning is experienced on landing?

- ☐ Apply nosewheel and main wheel braking alternately and abruptly.
- ☐ Apply full main wheel braking only.
- ☒ Apply aerodynamic braking to the fullest advantage.

17

If while in level flight, it becomes necessary to use an alternate source of static pressure vented inside the airplane, which of the following variations in instrument indications should the pilot expect?

- ☐ The vertical speed to momentarily show a descent.
- ☐ The vertical speed to show a climb.
- ☒ The altimeter to read higher than normal.

18

To level off at an airspeed higher than the descent speed, the addition of power should be made, assuming a 500 FPM rate of descent, at approximately

- ☒ 50 to 100 feet above the desired altitude.
- ☐ 150 to 200 feet above the desired altitude.
- ☐ 100 to 150 feet above the desired altitude.

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The best lift/drag ratio of a glider occurs when parasite drag is

☐ less than induced drag.

☐ equal to induced drag.

☒ equal to total drag.

20

(Refer to Figure 89.) Which of the following configurations will result in the shortest landing distance over a 50-foot obstacle to a wet runway?

See Attachments:

1. atp_89.jpg

figure 89

(Attachment
No.2)

- ☐ Brakes, spoilers, and reversers at 131,000 pounds gross weight.
- ☐ Brakes and reversers at 124,000 pounds gross weight.
- ☒ Brakes and spoilers at 122,500 pounds gross weight.

21

The coefficient of lift can be increased either by flap extension or by

- ☒ increasing the angle of attack.
- ☐ decreasing the 'nose-up' elevator trim setting.
- ☐ increasing the CAS.
- ☐ increasing the TAS.

22

On a dry runway the accelerate stop distance is increased

- ☐ by low outside air temperature.
- ☐ by uphill slope.
- ☒ by a lower take-off mass because the aeroplane accelerates faster to V1.
- ☐ by headwind.

23

When the outside air temperature increases, then

- ☐ the field length limited take-off mass increases but the climb limited take-off mass decreases.
- ☒ the field length limited take-off mass and the climb limited take-off mass increases.
- ☐ the field length limited take-off mass and the climb limited take-off mass decreases.
- ☐ the field length limited take-off mass decreases but the climb limited take-off mass increases.

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24

The climb gradient of an aircraft after take-off is 6% in standard atmosphere, no wind, at 0 ft pressure altitude. Using the following corrections: " $\pm 0,2 \%$ / 1 000 ft field elevation", " $\pm 0,1 \%$ / $^{\circ}\text{C}$ from standard temperature", " - 1 % with wing anti-ice" " - 0,5% with engine anti-ice", The climb gradient after take-off from an airport situated at 1 000 ft, 17 $^{\circ}\text{C}$; QNH 1013,25 hPa, with wing and engine anti-ice operating for a functional check is :

☐ 0.04

☐ 0.03

☒ 0.04

☐ 0.04

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With an true airspeed of 194 kt and a vertical speed of 1 000 ft/min, the climb gradient is about :

☐ 5°

☒ 0.03

☐ 3°

☐ 0.08

26

The maximum horizontal speed occurs when:

- ☐ The maximum thrust is equal to the total drag.
- ☐ The thrust does not increase further with increasing speed.
- ☒ The thrust is equal to minimum drag.
- ☐ The thrust is equal to the maximum drag.

27

For a piston engined aeroplane, the speed for maximum range is :

- ☐ that which gives the maximum lift to drag ratio.
- ☐ that which gives the maximum value of lift
- ☐ 1.4 times the stall speed in clean configuration.
- ☒ that which gives the minimum value of drag.

28

With regard to a unaccelerated horizontal flight, which of the following statement is correct?

- ☒ The minimum drag is is a function of the density altitude.
- ☐ The minimum drag is independant of the aircraft mass.
- ☐ The minimum drag is a function of the pressure altitude.
- ☐ The minimum drag is proportional to the aircraft mass.

29

The Density Altitude

- ☐ is used to establish minimum clearance of 2.000 feet over mountains.
- ☐ is used to calculate the FL above the Transition Altitude.
- ☐ is equal to the pressure altitude.
- ☒ is used to determine the aeroplane performance.

30

An aeroplane is in a power off glide at best gliding speed. If the pilot increases pitch attitude the glide distance:

- ☐ remains the same.
- ☒ decreases.
- ☐ increases.
- ☐ may increase or decrease depending on the aeroplane.

31

Which of the following statements regarding filing a flight plan is correct?

- ☐ A flight plan should be filed when a national FIR boundary will be crossed.
- ☒ A flight plan should be filled atleast one hour before departure
- ☐ A flying college can file repetitive flight plan for VFR flights.
- ☐ Any flight plan should be filed at least 10 minutes before departure.

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32

If an aircraft is consuming 91 pounds of fuel per hour and groundspeed is 168 knots, how much fuel is required to travel 457 NM?

☐ 265 pounds.

☐ 291 pounds.

☒ 248 pounds.

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33

If an aircraft has a transponder, encoding altimeter, and DME, the proper equipment suffix to be entered on a flight plan is

☒ A.

☐ U.

☐ R.

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34

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 121 NM.

☒ 176 NM.

☐ 215 NM.

35

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.
- ☒ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.

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36

From which of the following would you expect to find details of the Search and Rescue organisation and procedures (SAR) ?

- ☐ ATCC broadcasts
- ☒ SIGMET
- ☒ AIP (Aeronautical Information Publication)
- ☐ NOTAM

37

From which of the following would you expect to find information regarding known short unserviceability of VOR, TACAN, and NDB ?

☒ B AIP (Aeronautical Information Publication)

☐ D ATCC broadcasts

☐ A NOTAM

☐ C SIGMET

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38

You are flying a constant compass heading of 252° . Variation is 22°E , deviation is 3°W and your INS is showing a drift of 9° right. True track is ?

☒ 280°

☐ 242°

☐ 262°

☐ 224°

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39

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☒ B XXXX

☐ D ZZZZ

☐ A AAAA

☐ C IIII

40

A "current flight plan" is a:

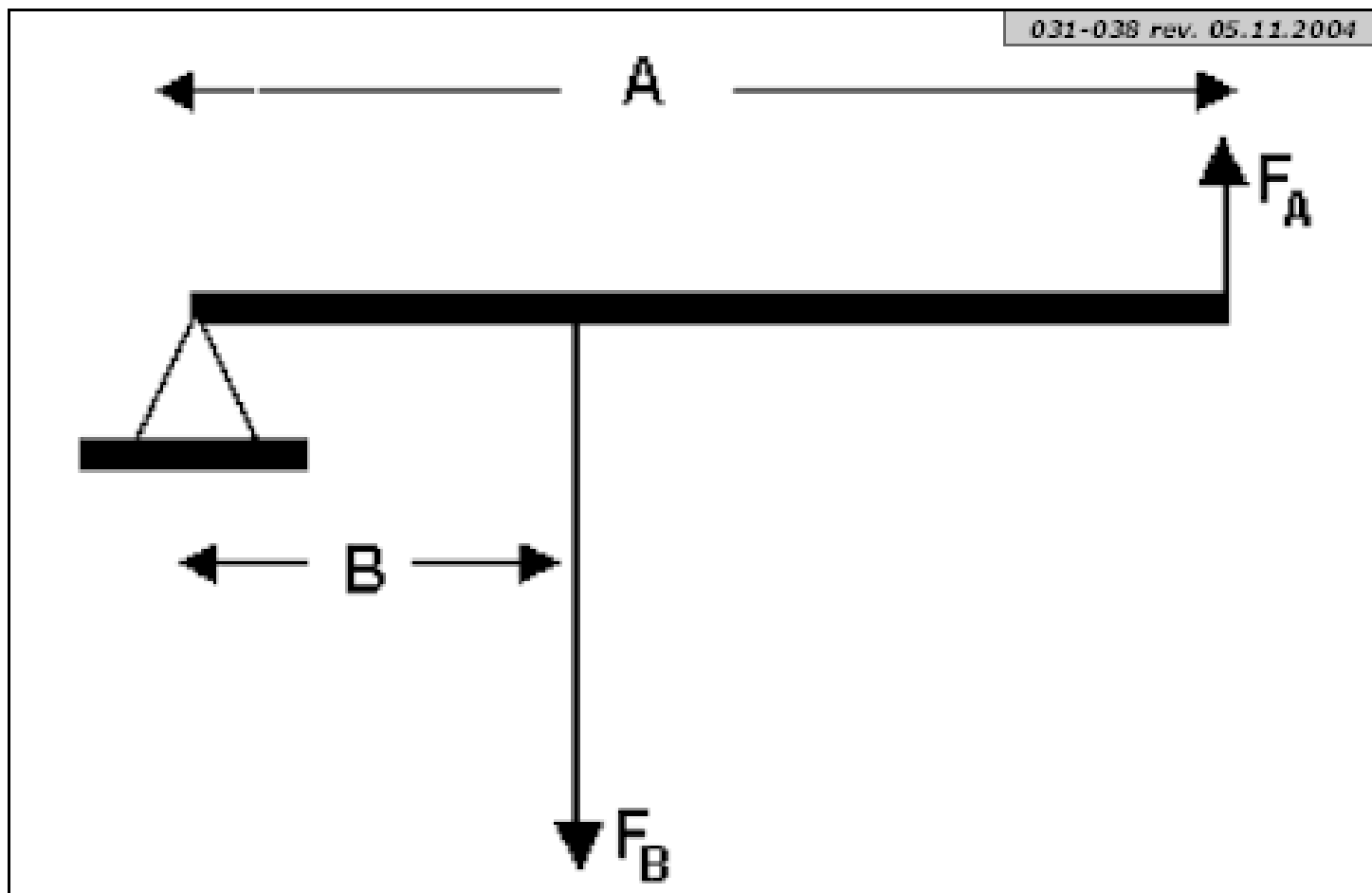
- ☐ *flight plan with the correct time of departure.*
- ☐ *filed flight plan.*
- ☐ *filed flight plan with amendments and clearance included.*
- ☒ *flight plan in the course of which radio communication should be practised between aeroplane and ATC.*

Attachment No. 1

Questions 15

Annex

031-038.gif

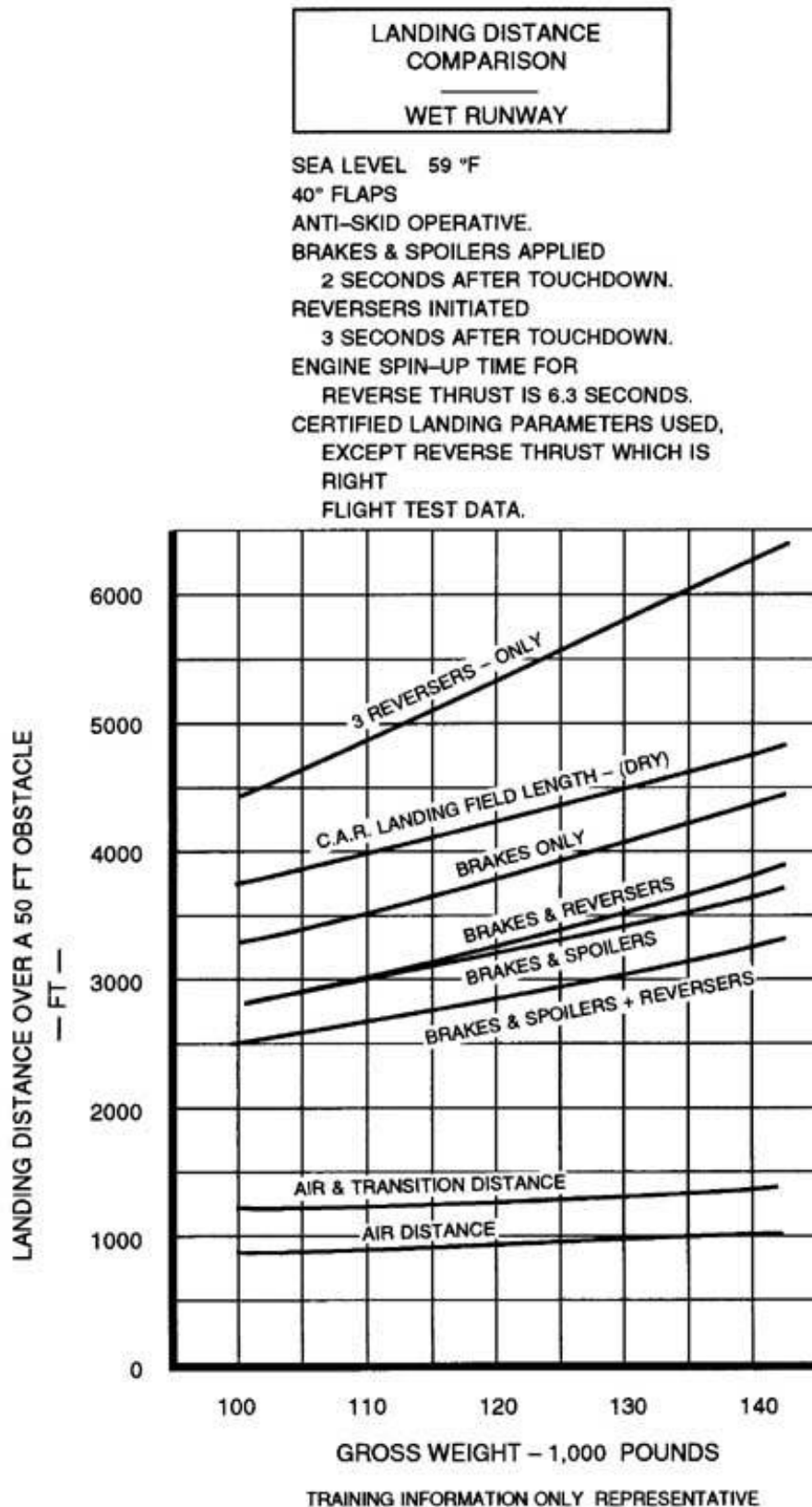


Attachment No. 2

Questions 20

figure 89

atp_89.jpg



NORMAL LANDING

Figure 89. B-727 – Normal Landing – Wet Runway © ASA